UNIVERSITÀ di VERONA
Dipartimento di INFORMATICA

Discrete Optimization and Decision Making

Integer Linear Programming - Modeling Techniques

Master's degrees in Data Science - Artificial Intelligence - Mathematics
Academic Year 2024/2025

Romeo Rizzi (Dept. of Computer Science, Univ. of Verona)
Roberto Zanotti (Dept. of Clinical and Experimental Sciences, Univ. of Brescia)

(Mixed) Integer Linear Programming: Modeling Techniques

Binary Choice

Most common use of binary variables in IP

- selection from a set
- choice between options
- activation/deactivation of components/parts/elements

Notable examples: **Knapsack Problems**

- Knapsack Problem
- Multidimensional Knapsack Problem
- Multiple-choice Knapsack Problem
- Multidimensional Multiple-choice Knapsack Problem

$\vdots$

The Multidimensional Multiple-choice Knapsack Problem

- $\mathcal{G} = \{G_1, G_2, \ldots, G_n\} \rightarrow$ collection of n disjoint group of items
- $n_i = |G_i|$, $i = 1, ..., n \rightarrow$ cardinality of each group
- $R = \{1, ..., m\} \rightarrow$ set of resources
- c_r, $r = 1, \ldots, m \rightarrow$ availability of each resource
- p_{ij}, $i = 1, ..., n$, $j = 1, ..., n_i \rightarrow$ profit
- w_{ij}^r, $i = 1, ..., n$, $j = 1, ..., n_i$, $r \in R \rightarrow$ resource consumption

$$\text{Maximize } z = \sum_{i=1}^{n} \sum_{j=1}^{n_i} p_{ij} x_{ij} \tag{1}$$

subject to:

$$\sum_{i=1}^{n} \sum_{j=1}^{n_i} w_{ij}^r x_{ij} \leq c_r \qquad r = 1, \ldots, m, \tag{2}$$

$$\sum_{j=1}^{n_i} x_{ij} = 1 \qquad i = 1, \ldots, n, \tag{3}$$

$$x_{ij} \in \{0, 1\} \qquad i = 1, \ldots, n, j = 1, \ldots, n_i. \tag{4}$$

Selection from a set

Given a set of binary variables x_j, each one associated with an item j belonging to a set T of finite dimension, we want to:

select at least one item:

$$\sum_{j \in T} x_j \geq 1,$$

select exactly one item:

$$\sum_{j \in T} x_j = 1,$$

select at most one item:

$$\sum_{j \in T} x_j \leq 1.$$

The Set Covering Problem

Application example: we want to cover an entire area with a connected system of stations to monitor air pollution. Each station covers part of the area. The goal is to minimize the number of stations to open, while covering the entire area.

Data:

T	$=$	$\{t_1, t_2, t_3, \ldots, t_m\}$ area divided into regions
t_i	$=$	i-th region
F	$=$	$\{F_1, F_2, \ldots, F_n : F_j \subseteq T\}$ monitoring stations
$F_j \subseteq T$	$=$	monitoring station j seen as a subset of T that covers one or more regions

Decision variables:

$$x_j = \begin{cases} 1 & \text{if I activate station } j, \\ 0 & \text{otherwise.} \end{cases}$$

Formulation:

$$min \sum_{j=1}^{n} x_j$$

$$\sum_{j=1}^{n} \delta_{ij} x_j \geq 1 \quad \forall i,$$

$$x_j \in \{0, 1\} \quad j = 1, ..., n$$

where parameter δ_{ij} is equal to 1 if region t_i is covered by station F_j, 0 otherwise.

Set Partitioning Problem

$$min \sum_{j=1}^{n} x_j$$

$$\sum_{j=1}^{n} \delta_{ij} x_j = 1 \quad \forall i,$$

$$x_j \in \{0, 1\} \quad j = 1, ..., n$$

Set Packing Problem

$$max \sum_{j=1}^{n} x_j$$

$$\sum_{j=1}^{n} \delta_{ij} x_j \leq 1 \quad \forall i,$$

$$x_j \in \{0, 1\} \quad j = 1, ..., n$$

Dependent decisions (I)

Decision A (binary variable x) can be taken ($x = 1$) if and only if decision B (binary variable y) has been taken ($y = 1$).

$$x \leq y$$

- if $y = 0$ (decision B not taken) $\Rightarrow x \leq 0 \Rightarrow x = 0$ (decision A cannot be taken);
- if $y = 1$ (decision B taken) $\Rightarrow x \leq 1 \Rightarrow x \in \{0, 1\}$ (decision A <u>can</u> be taken, but not necessarily).

Dependent decisions (II)

Variable $x \geq 0$ can take strictly positive values if and only if another decision (binary variable y) has been taken ($y = 1$):

$$x \leq My \qquad M \text{ large enough}$$

- if $y = 0$ (decision not taken) $\Rightarrow x \leq 0 \Rightarrow x = 0$ (x can only take value 0);
- if $y = 1$ (decision taken) $\Rightarrow x \leq M$ (x free, superiorly limited by M $\rightarrow$ choice of M is crucial to make the constraint meaningful)

Disjunctive constraints: case with two constraints

$\mathbf{x}$ vector of non-negative variables.

$$\mathbf{a}^T\mathbf{x} \geq b \quad (1)$$

$$\mathbf{c}^T\mathbf{x} \geq d \quad (2)$$

constraints where $\mathbf{a}$ and $\mathbf{c}$ are non-negative vectors. Only one of the two constraints can be active:

$$\mathbf{a}^T\mathbf{x} \geq yb$$

$$\mathbf{c}^T\mathbf{x} \geq (1-y)d$$

$$y \in \{0, 1\},$$

if $y = 1$, constraint (1) is active while constraint (2) is satisfied by all $\mathbf{x}$ ($\mathbf{c}^T\mathbf{x} \geq 0$) and is therefore redundant; vice-versa if $y = 0$.

Generalization to the case of more than 2 constraints I

Given m constraints $\mathbf{a}_i^T \mathbf{x} \geq b_i$, $i = 1, ..., m$ where $\mathbf{a}_i \geq 0$ for all i, the following formulation ensures that k constraints are satisfied:

$$\mathbf{a}_i^T \mathbf{x} \geq b_i y_i, \qquad i = 1, ..., m$$

$$\sum_{i=1}^{m} y_i = k,$$

$$y_i \in \{0, 1\} \quad i = 1, ..., m.$$

Generalization to the case of more than 2 constraints II

If we have m constraints $\mathbf{a}_i^T\mathbf{x} \geq b_i$, $i = 1, ..., m$ where the components of $\mathbf{a}_i$ can be ≤ 0, the formulation becomes:

$$\mathbf{a}_i^T\mathbf{x} \geq b_i - M_i(1 - y_i), \qquad i = 1, ..., m$$

$$\sum_{i=1}^{m} y_i = k,$$

$$y_i \in \{0, 1\} \quad i = 1, ..., m.$$

- if $y_i = 1$ the corresponding constraint is active;
- if $y_i = 0$ the corresponding constraint becomes $\mathbf{a}_i^T\mathbf{x} \geq b_i - M_i$ which, for a large enough $M_i >> 0$, is always satisfied and redundant.

Variables with discrete finite domain

Variable x can take only values from a finite set $A = \{a_1, a_2, ..., a_m\}$:

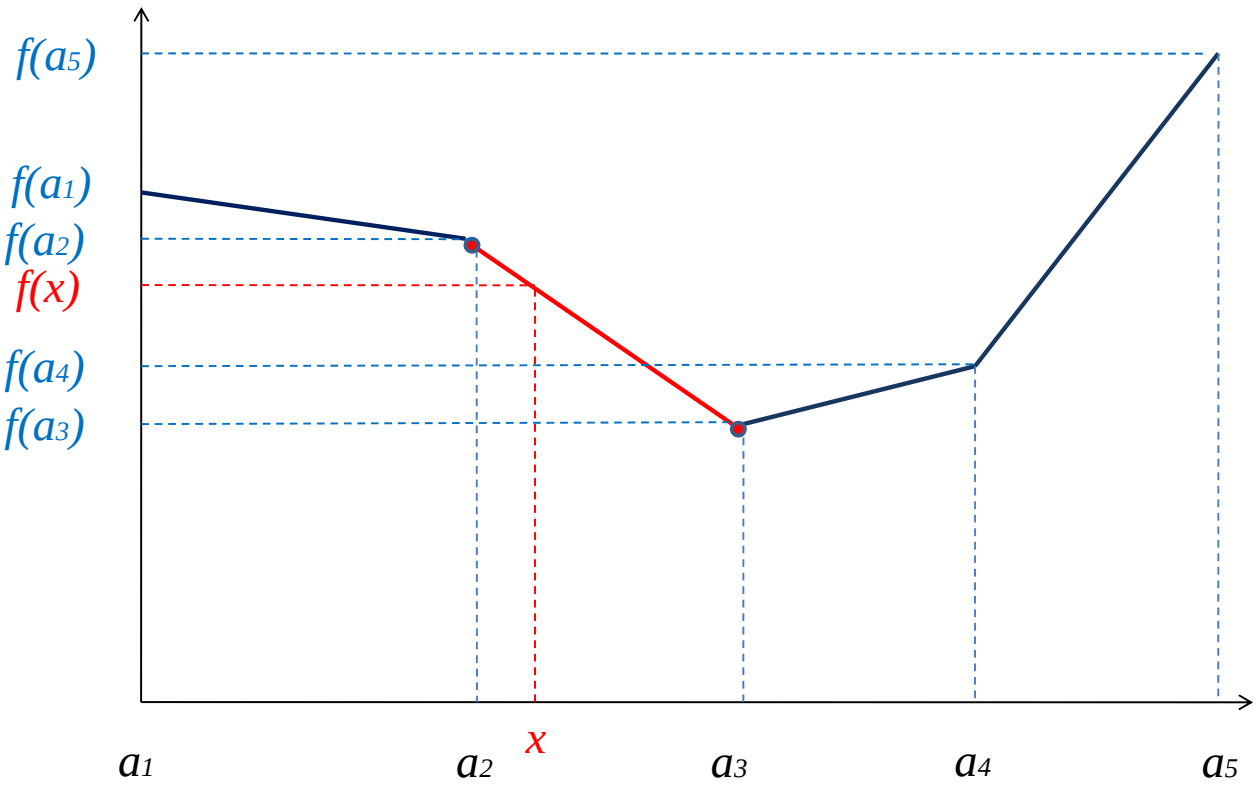
$$x = \sum_{j=1}^{m} a_j y_j$$

$$\sum_{j=1}^{m} y_j = 1$$

$$y_j \in \{0, 1\} \qquad \forall j = 1, 2, ..., m$$

This is called *Special Ordered Set* of type 1 (SOS1).

Piecewise-linear function



We want to model a piecewise linear function, i.e. a set of straight-line segments that connects a sequence of points, whose coordinates can be expressed as $(a_i, f(a_i)), (a_{i+1}, f(a_{i+1})), \ldots, (a_n, f(a_n))$.

Formulation I

$$f(x) = \sum_{i=1}^{n} \lambda_i f(a_i)$$

$$\sum_{i=1}^{n} \lambda_i = 1$$

$$\lambda_1 \leq y_1$$

$$\lambda_i \leq y_{i-1} + y_i, \quad i = 2, ..., n-1$$

$$\lambda_n \leq y_{n-1}$$

$$\sum_{i=1}^{n-1} y_i = 1$$

$$\lambda_i \geq 0, \quad i = 1, ..., n$$

$$y_i \in \{0, 1\}, \quad i = 1, ..., n-1.$$

Formulation II

$$f(x) = \sum_{i=1}^{n} \lambda_i f(a_i)$$

$$\sum_{i=1}^{n} \lambda_i = 1$$

$$\lambda_i \leq y_i \quad i = 1, ..., n$$

$$\sum_{i=1}^{n} y_i \leq 2$$

$$y_i + y_j \leq 1 \quad i = 1, ..., n-2, j = i+2, ..., n$$

$$y_i \in \{0, 1\} \quad i = 1, ..., n$$

Special Order Set of type 2 (SOS2): at most 2 binary variables in a set can be selected, and these two variables must be consecutive.

The Pollution Routing Problem

- $G = (\mathcal{N}, \mathcal{A})$ complete directed connected graph
- a set of nodes $\mathcal{N} = \{0, 1, 2, \ldots, n\}$ ($\mathcal{N}_0 = \mathcal{N} \setminus \{0\}$)
- a set of arcs $\mathcal{A} = \{(i, j) : i, j \in \mathcal{N}, i \neq j\}$
- a set $\mathcal{K} = \{1, 2, \ldots, m\}$ of homogeneous vehicles with capacity Q
- a demand q_i and a time window $[a_i, b_i]$, $\forall i \in \mathcal{N}_0$
- a travel distance d_{ij} associated with each arc $(i, j) \in \mathcal{A}$
- a service time t_i, $\forall i \in \mathcal{N}_0$

Objective: Find a set of routes (starting and ending in 0) to meet the demand of all customers that minimizes total cost (emission, operational, driver costs)

Fuel and CO_2 emissions

Comprehensive emission model

$$E(t, f) = \lambda(kNVt + w\gamma\alpha d + \gamma\alpha f d + \beta\gamma\frac{d^3}{t^2})$$

Variables

- t travel time
- f load

Parameters

- k, N, V engine parameters
- w, γ, β vehicle parameters
- d, α arc parameters
- λ fuel parameter

Decision Variables

Decision Variables:

- x_{ij}: 1 if a vehicle traverses $(i, j) \in \mathcal{A}$, 0 otherwise
- f_{ij}: amount of commodity flowing on arc $(i, j) \in \mathcal{A}$
- w_{ij}: time used by a vehicle to travel on arc $(i, j) \in \mathcal{A}$
- y_j: starting time of service at node $j \in \mathcal{N}_0$
- s_j: route duration if j is the last node visited before the depot

Constraints

$$\sum_{j \in \mathcal{N}} x_{0j} = m \tag{5}$$

$$\sum_{j \in \mathcal{N}} x_{ij} = 1 \quad \forall i \in \mathcal{N}_0 \tag{6}$$

$$\sum_{i \in \mathcal{N}} x_{ij} = 1 \quad \forall j \in \mathcal{N}_0 \tag{7}$$

$$\sum_{j \in \mathcal{N}} f_{ji} - \sum_{j \in \mathcal{N}} f_{ij} = q_i \quad \forall i \in \mathcal{N}_0 \tag{8}$$

$$q_j x_{ij} \leq f_{ij} \leq (Q - q_i) x_{ij} \quad \forall (i,j) \in \mathcal{A} \tag{9}$$

$$a_i \leq y_i \leq b_i \quad \forall i \in \mathcal{N}_0 \tag{10}$$

$$y_i - y_j + t_i + w_{ij} \leq M_{ij}(1 - x_{ij}) \quad \forall (i,j) \in \mathcal{A} \tag{11}$$

$$y_j + t_j - s_j + w_{j0} \leq L(1 - x_{j0}) \quad \forall j \in \mathcal{N}_0 \tag{12}$$

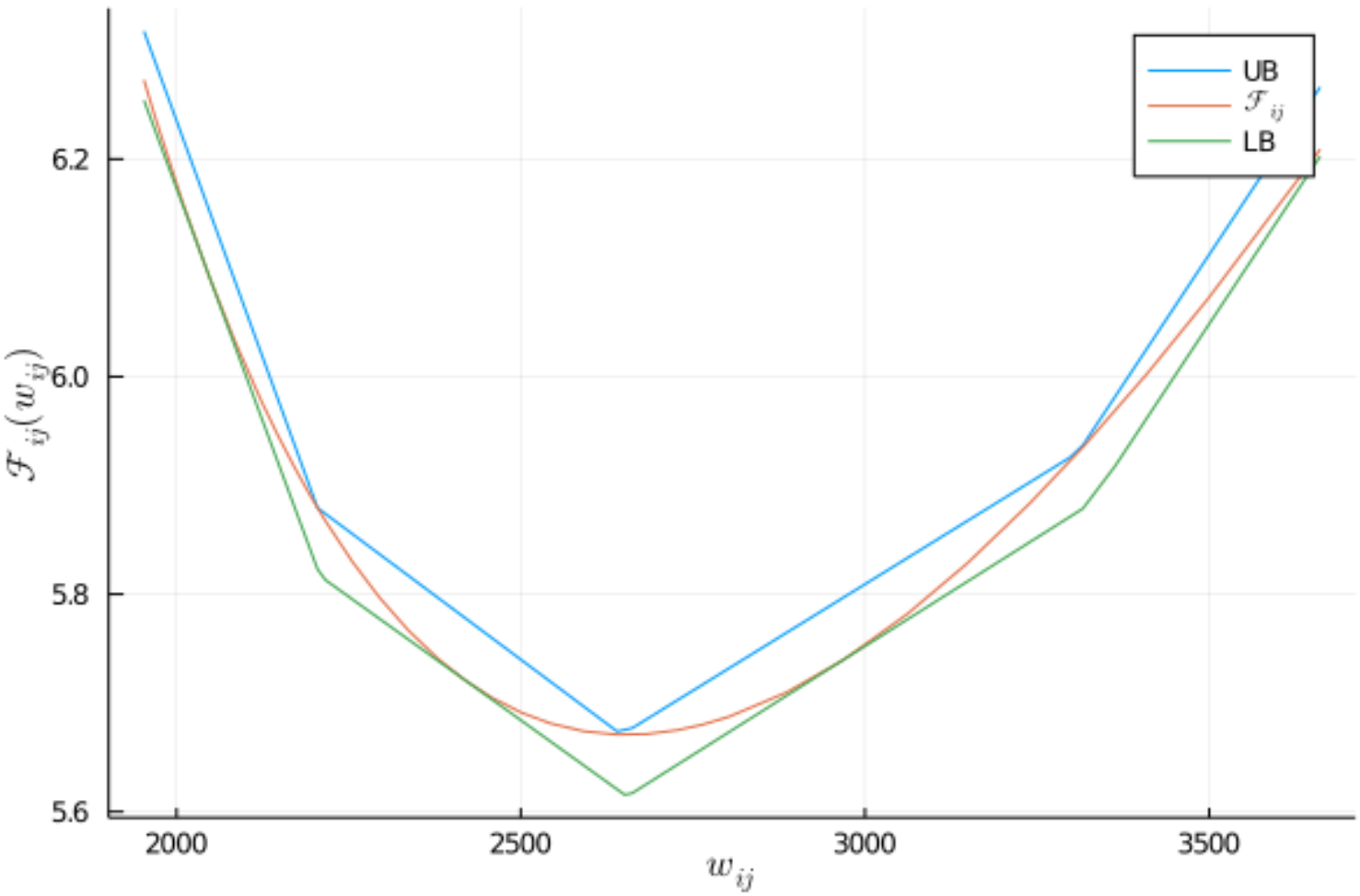
$$\frac{d_{ij}}{u_{ij}} x_{ij} \leq w_{ij} \leq \frac{d_{ij}}{l_{ij}} x_{ij} \quad \forall (i,j) \in \mathcal{A} \tag{13}$$

$$x_{ij} \in \{0,1\}, f_{ij} \geq 0 \quad \forall (i,j) \in \mathcal{A} \tag{14}$$

Objective Function (I)

$$
\begin{aligned}
\min \quad & p_1 \gamma \lambda \sum_{(i,j)\in\mathcal{A}} \alpha_{ij} d_{ij} f_{ij} && (15)\\
& + p_1 w \lambda \gamma \sum_{(i,j)\in\mathcal{A}} \alpha_{ij} d_{ij} x_{ij} && (16)\\
& + p_2 \sum_{j\in\mathcal{N}_0} s_j && (17)\\
& + p_1 \lambda k N V \sum_{(i,j)\in\mathcal{A}} w_{ij} && (18)\\
& + p_1 \lambda \gamma \beta \sum_{(i,j)\in\mathcal{A}} d_{ij}^3 / w_{ij}^2 && (19)
\end{aligned}
$$

Objective Function (II)



Piecewise approximation 1/3

$$\min \quad (15) + (16) + (17) + \sum_{(i,j)\in\mathcal{A}} \mathcal{F}_{ij}(w_{ij}) \tag{20}$$

$$\mathcal{F}_{ij}(w_{ij}) := p_1 \lambda k N V w_{ij} + p_1 \lambda \gamma \beta d_{ij}^3 / w_{ij}^2 \tag{21}$$

$$\downarrow$$

approximated by a piecewise linear function

Piecewise approximation 2/3

- $\mathcal{P} = \{1, 2, \ldots, p_{max}\}$ ordered set of breakpoints
- $F_{ij}^{p} = \mathcal{F}_{ij}(w_{ij}^{p}) \quad \forall p \in \mathcal{P}$
- θ_{ij}^{p} and ψ_{ij}^{p} are auxiliary variables for the piecewise formulation.

$$(PRP_{pwl}) \min \quad (15) + (16) + (17) + \sum_{(i,j)\in\mathcal{A}} \left(\sum_{p=1}^{p_{max}} F_{ij}^{p} \theta_{ij}^{p} \right) \tag{22}$$

Piecewise approximation 3/3

Additional constraints $\forall (i,j) \in \mathcal{A}$

$$\sum_{p=1}^{p_{max}} \theta_{ij}^{p} = x_{ij} \tag{23}$$

$$\sum_{p=1}^{p_{max}} w_{ij}^{p} \theta_{ij}^{p} = w_{ij} \tag{24}$$

$$\theta_{ij}^{1} \leq \psi_{ij}^{1} \tag{25}$$

$$\theta_{ij}^{p} \leq \psi_{ij}^{p-1} + \psi_{ij}^{p} \quad \forall p = 2, \ldots, p_{max} - 1 \tag{26}$$

$$\theta_{ij}^{p_{max}} \leq \psi_{ij}^{p_{max}-1} \tag{27}$$

$$\sum_{p=1}^{p_{max}-1} \psi_{ij}^{p} = 1 \tag{28}$$

$$\theta_{ij}^{p} \geq 0 \quad \forall p = 1, \ldots, p_{max} \tag{29}$$

$$\psi_{ij}^{p} \in \{0, 1\} \quad \forall p = 1, \ldots, p_{max} - 1. \tag{30}$$